

Kyra Veprek | Curriculum Vitae

kyraveprek24@gmail.com

Education

Brown University - Providence, RI 2021-2026
Ph.D., Cognitive Science (Expected)

Dissertation: “The Dynamics of Visual Perception and Locomotion in Human Crowds”

Brown University - Providence, RI 2021-2023
M.Sc., Cognitive Science

Thesis: “Walking Through a Crowd: Modeling Pedestrian Collision Avoidance Behavior”

University of Massachusetts, Amherst - Amherst, MA 2018-2021
B.A., Psychology; B.A. Studio Art

Honors Thesis: “Investigating the Horizontal Advantage in Symmetry Perception”

Peer-Reviewed Conference Proceedings

Feldmann, S., Veprek, K., & Warren, W. H. (2025). Formation of lanes and stripes in crossing pedestrian flows using an empirical human model. In *Proceedings of the Traffic and Granular Flow Conference. EPJ Web of Conferences* (Vol. 334, p. 04011). EDP Sciences.

Conference Presentations & Abstracts

Feldmann, S., Veprek, K., & Warren, W. (2025). Simulating crossing pedestrian flows with a vision-based model of collision avoidance. Talk presented at the *International Conference on Pedestrian and Evacuation Dynamics (PED)*, Prague, Czech Republic.

Veprek, K., Feldmann, S., & Warren, W. (2025). Visual model of locomotion reproduces lanes and stripes in crossing human crowds. *Journal of Vision*, 25(9), 2681-2681.

Veprek, K., & Warren W. (2024). How many potential collisions do we respond to at once: Investigating the efficacy of visual thresholds in human crowds. Poster presented at the *International Conference on Perception and Action*, Trondheim, Norway.

Veprek, K., & Warren, W. (2024). How many moving obstacles do we respond to at once? A temporal threshold model best accounts for collision avoidance in a crowd. *Journal of Vision*, 24(10), 1313-1313.

Veprek, K., & Warren W. (2023). Human collision avoidance with multiple moving obstacles. Talk presented as part of the symposium *Flow in Flight* at the *International Conference on Perception and Action*, Guadalajara, Mexico.

Veprek, K., & Warren, W. (2023). Visual Collision Avoidance in a Crowd. *Journal of Vision*, 23(9), 5545-5545.

Veprek, K., Esiason, J., and Zarch, R. (2021). The footprint of RPPforCS: A national look at communities served by research-practitioner-partnerships in computer science. Poster presented for *Research on equity and sustained participation in engineering, computing, and technology (RESPECT) IEEE*, Remote.

Workshops & Certifications

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| 2024 | Teaching Certificate II: Course Design
<i>The Harriet W. Sheridan Center for Teaching and Learning, Brown University</i> |
| 2024 | Workshop on Interacting Particle Systems: Analysis, Control, Learning, and Computation
<i>Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University</i> |
| 2022 | Teaching Certificate I: Reflective Teaching
<i>The Harriet W. Sheridan Center for Teaching and Learning, Brown University</i> |

Teaching

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| 2025 | Research Methods , Graduate Teaching Assistant
<i>Department of Cognitive & Psychological Sciences, Brown University</i> |
| 2024, 2023 | Introduction to Statistical Methods , Graduate Teaching Assistant
<i>Department of Cognitive & Psychological Sciences, Brown University</i> |
| 2023 | Learning & Conditioning , Graduate Teaching Assistant
<i>Department of Cognitive & Psychological Sciences, Brown University</i> |

Service & Leadership

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| 2024-2025 | Member , Diversity and Inclusion Action Plan, Community Outreach Committee
<i>Department of Cognitive & Psychological Sciences, Brown University</i> |
| 2024 | Lead Organizer , Graduate Program Recruitment Preview Event
<i>Department of Cognitive & Psychological Sciences, Brown University</i> |

- 2023-2025 **Lead Organizer,** Vision Science Reading Group
Department of Cognitive & Psychological Sciences, Brown University
- 2025, 2023 **Cognitive Science Representative**
College Day at Brown
- 2022-2023 **Graduate Student Representative**
Department of Cognitive & Psychological Sciences, Brown University
- 2022-2024 **Department Peer Mentor**
Department of Cognitive & Psychological Sciences, Brown University

Professional Societies

Vision Sciences Society

International Society for Ecological Psychology